

Judge report — C7 panel (2026-08-14, morning of upload)

Materials judged: deck/trackb.pdf (32 pp), deck/slides_content.md, deck/script.md, deck/qa_crib.md, README.md, the participant notebook, deck/*.png, figures/*.png. Nothing else.

J1 — rubric bureaucrat

criterion	weight	score	one-line justification
Originality & Uniqueness	15	11	Track B completion (sum/difference readout), the phase-trap discovery, and pre-registered hardware falsification are genuinely unusual; the core protocol is the PRL's and AQC is an off-the-shelf add-on.
Usefulness & Complexity	25	19	Quickstart reproduces the headline table from saved data; scripts re-run every hardware number; but the flagship compilation win is demonstrated NOT to function on present hardware, and the tooling is benchmark-specific.
Quantum Community Benefit	25	20	The ledger discipline (R### on every number), bug log with retractions, study notes and handover are teachable artefacts beyond the result itself; reuse of the Track-B certifier is plausible.
Presentation	35	24	Takeaway-sentence titles, scoreboard opener, negatives given equal billing — strong. Docked hard for: THREE TBD-NAMEs still on the title and team slides; script timing 10:20 against a stated 10:00 before any cut; speaker roster unassigned so "did every member speak" is UNVERIFIABLE; README quickstart's first line does not run (J3).
Total	100	74	

Where a median team beats this one: logistics. A median team has real names on the title slide, a first command that runs verbatim, and a talk that fits its slot without a cut list. This team has better science and worse housekeeping; the rubric weights housekeeping at 35%.

Banned-language sweep: no "quantum advantage" found. "free error bar" (S13/p.21) is costed in caption and script ("no additional circuits", diagnostic-only, R044) — PASS, borderline; keep the costing sentence adjacent to the word "free" wherever it appears. No causal single-run device comparisons found on slides (single-run status is disclosed). "Hardware" language is consistent with hardware having actually run.

J2 — physics professor

- THE weakest claim (see below): ϵ_{eff} is presented as measured; it is a bound.** A7 extracts $\epsilon_{\text{eff}} = -\ln(0.026)/576 = 6.3 \times 10^{-3}$. At 4,000 shots/quadrature, the per-quadrature σ is $1/\sqrt{4000} \approx 0.0158$, so a *fully dead* circuit yields $E|\chi| = \sigma\sqrt{(\pi/2)} \approx 0.020$ from estimator bias alone. Measured 0.026–0.033 is 1.3–1.7 $\times$ the pure-noise mean — consistent with noise floor. Hence 6.3×10^{-3} is a LOWER bound on ϵ_{eff} , not a measurement; the true error may be worse. The conclusion (win invisible on hardware) survives — indeed strengthens — but the wording "extracted from data" over-claims.
- $\langle Q \rangle_W - \langle Q \rangle_U$ deviations carry no uncertainties on the slide** (–0.04/–0.39/–0.25/–0.34, S13). House rule is "every number an uncertainty"; single-run, no bootstrap is disclosed in the ledger but not visible on the slide. A judge reading only slides sees naked numbers.
- The identity $\langle Q \rangle_W - \langle Q \rangle_U \equiv 0$ is correctly argued for Trotter factors (each term conserves Q); the slide's "for any W in this family" is right but "family" is undefined on-slide — one clause would close it.
- Conventions, unbiasedness, 3^w variance: internally consistent; the 2% variance-model check and the 12/12-seed table are the strongest physics-hygiene artefacts here. The non-

degenerate-by-design benchmark dodges the degeneracy question legitimately (and says so).

5. The miami 22.6σ statement (trap slide) is a single run used to illustrate a metric failure, not a device claim — acceptable as framed, fragile if a speaker improvises it into a device comparison. Script wording is safe; enforce it.

J3 — engineering pragmatist (executed, timed)

Fresh clone, README followed verbatim:

step	README says	result	time
1	git clone <this-repo> && cd 2026	FAILS twice: <this-repo> is a literal placeholder; after substituting the URL, cd 2026 fails — the directory is Qiskit-Hackathon-2026	clone 5.9 s
2	venv + pinned pip	works	15.2 s (warm wheel cache; expect minutes on a judge's cold machine)
3	headline table from saved data	works, prints the 12/12 table matching the README	0.013 s
—	notebook evidence without executing	34/98 cells carry embedded outputs; 56 embedded "PASS" strings visible without running anything	—
—	slide-QR target	HTTP 200	—

What breaks first: the first line of the quickstart. Everything after it works and is fast. Total honest time after fixing line 1: **~25 s warm, plausibly 3-5 min cold.** The "5-minute" claim is defensible only once line 1 is fixed.

12 questions, ordered by hostility (answer the panel would accept)

1. **"Your ϵ_{eff} of 6×10^{-3} — that's just the noise floor of $|x|$, isn't it?"** Accept: "Correct — 0.026 is near the $|\cdot|$ bias floor at these shots, so 6×10^{-3} is a lower bound on the effective error and the visibility gap is AT LEAST $3 \times$. The verdict is unchanged; the label should say 'bound'."
2. **"Slide 13's charge deviations have no error bars. Why should I trust them?"** Accept: single run, disclosed in R044; the per-quadrature sem is ~ 0.016 so -0.39 is many σ from zero; slides should carry that sem.
3. **"Your first README command doesn't run."** Accept: acknowledged; the fix is one line; everything downstream executes in seconds (demonstrate).
4. **"Three TBD names on the title slide the morning of judging?"** Accept only: the filled slide.
5. **"36 $\times$ on paper, 12 $\times$ wrong on hardware — why is the headline not simply 'we were wrong'?"** Accept: both are pre-registered claim types; slide 2 leads with the failure at equal size; the compilation number is deterministic and reproducible from the repo.
6. **"Why $n=6$ when your headline is $n=7$?"** Accept: the A7 arithmetic (control-arm cost $\times 4$, contrast saturated) — it is on a slide.
7. **"The echo beats you 27 $\times$. Why does your instrument exist?"** Accept: phase
8. per-observable profile, which no scalar test returns; stated with the loss, not instead of it.
9. **"Process infidelity 0.96 — your compiled gate is not U at all."** Accept: compressed against one input by design; legitimate for a Hadamard test only; the slide says exactly this.
10. **"Is the 22.6σ miami number a device comparison from one run?"** Accept: no — a metric-failure exhibit; survival ordering across models is shown inconsistent, no device ranking claimed.

11. **"Did every member speak?"** Accept only: yes, with named blocks. Script has A/B/C blocks but no names attached as of judging morning.
 12. **"What's actually new versus the PRL?"** Accept: the sum/difference completion, the trap + one-gate fix, pre-registered hardware falsification, and the boundary maps; the base protocol is credited to the paper on the title slide.
 13. **"First open implementation — verified how?"** Accept: date-stamped search scope in the repo, plus the rehearsed concession if prior art surfaces.
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Singular findings

THE weakest claim: A7 / R054 wording that $\epsilon_{\text{eff}} \approx 6 \times 10^{-3}$ is "extracted from data." It is a lower bound extracted from a value consistent with the $|\chi|$ noise floor. *30-minute fix (belongs to /c6):* on A7 and in the script, change "extracted" to "bounded below by"; add one line "measured $|\chi| \approx$ noise floor 0.020 at these shots $\Rightarrow \epsilon_{\text{eff}} \geq 6 \times 10^{-3}$, gap to visibility $\geq 3 \times$." Conclusion unchanged, over-claim removed.

THE weakest slide: the Team slide (S17) — three TBD-NAMES and no who-built-what, on a rubric that explicitly scores whether every member spoke, under the criterion worth 35%. *30-minute fix (belongs to humans + /c6):* fill three names, one line of ownership each, map to script blocks A/B/C.

Timing check: script totals 10:20 against a stated 10:00 slot — DOES NOT FIT until cut 1 is applied; with cut 1 it is 10:00 with zero slack, violating the script's own 10% slack rule.

Speaking check: blocks exist for three speakers; UNVERIFIABLE against a roster that does not exist in the materials.

No further comment. Fixes belong to /c5 and /c6.